Q(1)	Q(2)	Q(3)	Q(4)	Q(5)	Q(6)	Q(7)	Q(8)

Q(1) If $y = \tan^{-1}(\ln x)$, then $\frac{dy}{dx}|_{x=1} =$

a) -1

b) 1

c) 2

d) 0

Q(2) If $y = (x)^{\sinh x}$, then $\frac{dy}{dx}|_{x=1} =$

a) $\sinh(1)$

b) $\cosh(1)$

c) $\sinh(1) + \cosh(1)$

d) 0

Q(3) $\int_0^{\pi} \tan^4 x \, dx =$

a) 2π

b) 3π

c) 4π

d) π

Q(4) $\lim_{x \to 0^+} (\ln(\sinh x) - \ln(\sin^{-1} x)) =$

a) $\ln 2$

b) e

c) 1

d) 0

Q(5) $\int x\sqrt{9 - x^2} \, dx =$

a) $27 \int \tan\theta \sec^3\theta \, d\theta$

b) $9 \int \sin\theta \cos^2\theta \, d\theta$

c) $27 \int \sin\theta \cos^2\theta \, d\theta$

d) $3 \int \sin\theta \cos^2\theta \, d\theta$

Q(6) $\int \sin^6 x \cos^3 x \, dx =$

a) $\frac{1}{9}\sin^9 x - \frac{1}{7}\sin^7 x + c$

b) $\frac{1}{6}\sin^6 x - \frac{1}{8}\sin^8 x + c$

c) $\frac{1}{7}\sin^7 x - \frac{1}{9}\sin^9 x + c$

d) $\frac{1}{5}\sin^5 x - \frac{1}{7}\sin^7 x + c$

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Q(7) $\int x.2^x\,dx =$

a) $\dfrac{x.2^x}{(\ln 2)^2} - \dfrac{2^x}{\ln 2} + c$

b) $\dfrac{x.2^x}{\ln 2} - \dfrac{2^x}{(\ln 2)^2} + c$

c) $\dfrac{2^x}{\ln 2} - \dfrac{2^x}{(\ln 2)^2} + c$

d) $\dfrac{2^x}{\ln 2} - \dfrac{x.2^x}{(\ln 2)^2} + c$

Q(8) $\tan^2(\sin^{-1} x) =$

a) $\dfrac{x^2}{1-x^2}$

b) $\dfrac{1-x^2}{x^2}$

c) $\dfrac{x}{\sqrt{1-x^2}}$

d) $\dfrac{x^2}{x^2-1}$

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For equestions (9) – (12): Write T (True) or F (False).

Q(9) If $\log_x(2) = 3$, then $x = \sqrt[3]{2}$. ( )

Q(10) If $(e)^{e^{2x-1}} = 5$, then $x = \dfrac{1+\ln(\ln 5)}{2}$. ( )

Q(11) If $\lim\limits_{x\to 0}\dfrac{a^x-1}{x}$ exists, then $a \neq 0$. ( )

Q(12) The equation $\cosh x = \sinh x$ has a solution. ( )

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